

$$\omega = 2\pi f$$

$$Q1. \quad Y = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{-2526.54 + \frac{432}{25}\pi j} = \sqrt{2527.12478.77} = 50.27 \angle 89.38^\circ$$

$$= 0.54 + 50.17j$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{2499.51 \angle 1.05^\circ}$$

$$= 50.0 \angle 0.525^\circ = 50 + 0.46j \, \Omega$$

$$R = 0.54 \quad N_p/m \Rightarrow \left| 20 \log(e^{-\alpha L}) \right| : 144 \text{ dB}$$

$$\alpha L \cdot (20 \log e)$$

$$Q2. \quad (i) \quad \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-25 + 50j}{75 + 50j} = 0.077 + 0.615j$$

$$\textcircled{O} \quad Z_{in} = Z_0 \cdot \frac{Z_L + jZ_0 \tan \beta L}{Z_0 + jZ_L \tan \beta L}$$

$$(ii) \quad Z_{in} = Z_0 \cdot \frac{Z_0}{Z_L} = \frac{Z_0^2}{Z_L} = \frac{50^2}{(25 + 50j)} = 20 - 40j$$

$$(iii) \quad \Gamma(-L) = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{20 - 40j - 50}{20 - 40j + 50} = \frac{-30 - 40j}{70 - 40j} = -\frac{1}{13} - \frac{8}{13}j$$

$$\approx -0.077 - 0.615j$$

$$\approx 0.62 \angle -97.13^\circ$$

$$(iv) \quad VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.61}{1 - 0.62} = 4.26 = 12.54 \text{ dB}$$

$$(v) \quad Z_{in} = Z_0 = 50 \, \Omega$$

$$Q3. \quad (a) \quad \text{From } \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L - 60}{Z_L + 60} = 0.4 \angle 60^\circ \Rightarrow Z_L = 66.32 + 54.70j \, \Omega$$

$$(Z_L - 60) = 0.4 \angle 60^\circ (Z_L + 60)$$

$$= 0.4 Z_L \angle 60^\circ + 24 \angle 60^\circ$$

$$Z_L (1 - 0.4 \angle 60^\circ) = 60 + 24 \angle 60^\circ$$

$$\Rightarrow Z_L = \frac{60 + 24 \angle 60^\circ}{1 - 0.4 \angle 60^\circ} = 66.32 + 54.70j \, \Omega$$

$$(b) \quad \Gamma_{in} = \Gamma \cdot e^{-2j\beta L} = 0.4 \angle 60^\circ \cdot e^{-2j \cdot 0.3 \cdot 2\pi} = -0.365 - 0.163j = 0.4 \angle -156^\circ$$

$$(c) \quad Z_{in} = \frac{1 + \Gamma_{in}}{1 - \Gamma_{in}} = \frac{1 + 0.4 \angle -156^\circ}{1 - 0.4 \angle -156^\circ} = 26.65 - 10.33j \, \Omega$$

Q4. $SWR = \frac{1+|\Gamma_L|}{1-|\Gamma_L|} = \frac{3}{1} \Rightarrow |\Gamma_L| = 0.2$

$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \pm 0.2 = \frac{100 - Z_0}{100 + Z_0} \Rightarrow Z_0 = 66.7 \Omega \approx 150 \Omega$

$V_1^+ + V_1^- + V_2^+ = V_1^+ + \Gamma_L V_1^+ + \Gamma_L \Gamma_S V_1^+$
 $= (1 + \Gamma_L + \Gamma_L \Gamma_S) V_1^+ = 3V$

$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L - 50}{Z_L + 50} \Rightarrow 1 + \frac{Z_L - 50}{Z_L + 50} = 0.6$

$\Gamma_S = \frac{Z_S - Z_0}{Z_S + Z_0} = \frac{50 - 50}{50 + 50} = 0 \quad \frac{Z_L - 50}{Z_L + 50} = -0.4$

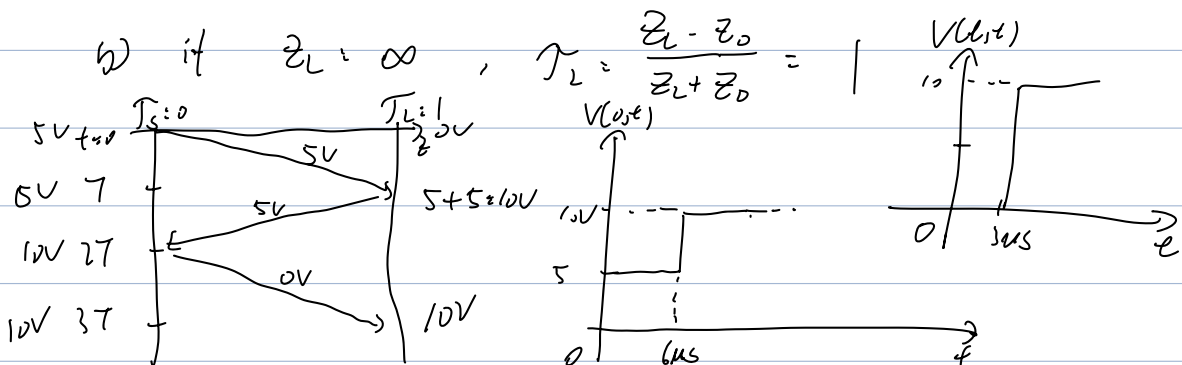
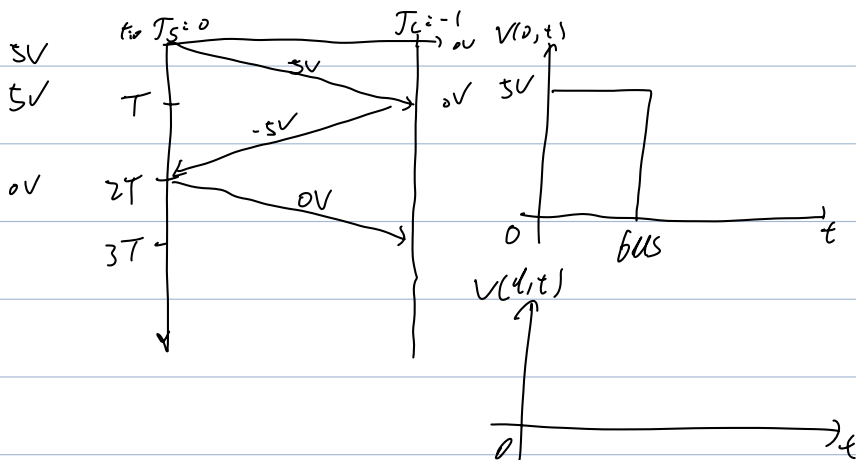
$Z_L - 50 = -0.4 Z_L - 20$

$1.4 Z_L = 30$

$Z_L = \frac{30}{1.4} \approx 21.43 \Omega$

Q8. a) if $Z_L = 0$, then $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - 50}{0 + 50} = -1$

$\Gamma_S = 0$

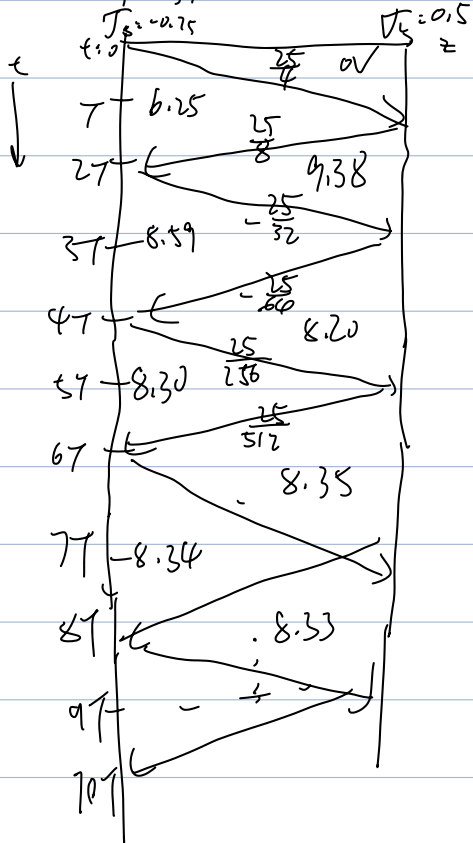


$$Q2. J_0 = \frac{z_s - z_0}{z_s + z_0} = \frac{30 - 50}{30 + 50} = -0.25$$

$$J_L = \frac{z_L - z_0}{z_L + z_0} = \frac{150 - 50}{150 + 50} = 0.5$$

$$V^+ = \frac{z_0}{z_s + z_0} V_s = \frac{50}{30 + 50} \cdot 10 = \frac{25}{4} V = 6.25 V$$

$$T = \frac{0.3}{3 \times 10^8} = 1 ns$$



$$\text{At steady state, } V(z, \infty) = \left(\frac{z_L}{z_L + z_s} \right) V_s = 8.33 V$$

$$T = 1 ns$$

$$Q5. (a) V(0,0) = \frac{z_0}{R_0 + z_0} V_s = 6V$$

$$\Rightarrow \frac{50}{R_0 + 50} = 0.4$$

$$50 = 0.4 R_0 + 20$$

$$30 \times \frac{5}{2} = 75$$

$$R_0 = 75 \Omega$$

$$(b) J_s = \frac{75 - 50}{75 + 50} = 0.2$$

$$V_1^+ = 6V \quad V_1^- = 2V$$

$$J_L = \frac{100 - 50}{100 + 50} = \frac{1}{3} \approx 0.33$$

$$V_2^+ = 0.4V$$

$$8.4V$$

$$c) \quad 3ms \times 10^4 = \underline{\underline{300m}}$$

$$V_1^+ + \Gamma_L V_1^+ + \Gamma_L \Gamma_S V_1^+ = 3$$

$$1 + \Gamma_L + \Gamma_L \cdot \frac{1}{5} = \frac{1}{2} \Rightarrow \Gamma_L = -\frac{1}{2} \cdot \frac{5}{2} = -\frac{5}{2}$$

$$\Gamma_L = \frac{z_f - z_0}{z_f + z_0} = -\frac{5}{12} \Rightarrow z_f = 20.5 \Omega$$